DNN Programming Assignment-2: Report

Variant: seeds 11, 22, 33 | hidden size 64 | student-ID last digit 9 (odd: LSTM forget-gate bias ablation) | PyTorch, GPU (CUDA)

Assumptions

P1: RNN/LSTM/GRU share Adam lr 1e-3, batch 128, gradient-norm clip 1.0 and equal epochs (HAR 20, adding 30). The epoch with best validation score (HAR macro-F1, adding MSE) is evaluated on test. HAR validation = 4 of 21 training subjects per seed, normalisation from training part only. nn.RNN/LSTM/GRU are used for training; from-scratch cells for tests and the gradient analysis. P2: the instructor's 3-token example was not available, so a hand-checkable example is used (code and hand calculation agree). Task A: patch 8 (16 patches + CLS), 2 pre-LN encoder layers, d=128, 4 heads, AdamW 5e-4, 20 epochs, selection on validation macro-F1. Task B: 4 layers, d=128, context 128, AdamW 1e-3, 100-step warm-up + cosine decay, 12x250 steps (ablations 4x250), validation on the whole last 10% in non-overlapping 128-character windows. nn.MultiheadAttention / nn.Transformer* are not used.

Problem 1 - one-page analysis

(1) Memory (gradient plot, T=100). Ratio of $||dL/dh_t||$ at t=1 to t=100, before / after training: RNN 4.92e-05 / 1.15e-04; LSTM 4.23e-09 / 2.94e-01; GRU 9.54e-20 / 6.01e-01. At initialisation all three vanish, the GRU most (1e-19). After training the LSTM and GRU gradients are almost flat across time (ratio 0.29 and 0.60), i.e. the gates learned to carry error signal back to early steps; the RNN stays at ~1e-4, so its early steps receive almost no gradient.

(2) Where each model fails as T grows. Solved = test MSE < 0.05 (constant baseline 0.170).

T	RNN MSE (solved)	LSTM MSE (solved)	GRU MSE (solved)
50	0.0241 (3/3)	0.0046 (3/3)	0.0015 (3/3)
100	0.1225 (0/3)	0.0662 (1/3)	0.0021 (3/3)
200	0.1678 (0/3)	0.1684 (0/3)	0.0031 (3/3)
400	0.1538 (0/3)	0.1535 (0/3)	0.0027 (3/3)

The RNN fails already at T=100, the LSTM at T=200 (stuck at baseline level for all 30 epochs), while the GRU solves every length with MSE 0.0015-0.0031. In the T=200 learning curves the GRU leaves the plateau at about epoch 13; LSTM and RNN never do within 30 epochs. Ablation (T=200): LSTM forget-bias 0 gives 0.1685 +- 0.0002 and bias 1 gives 0.1684 +- 0.0004, both 0/3 solved - the bias only changes the first few validation epochs (bias 0 is higher early) and does not rescue training in this budget.

(3) Accuracy - parameters - time (HAR, hidden 64).

Model	Test acc	Macro-F1	Params	s/epoch
RNN	0.8422 +- 0.0116	0.8388	5,190	0.214
LSTM	0.8973 +- 0.0221	0.8973	19,590	0.206
GRU	0.8896 +- 0.0046	0.8888	14,790	0.179

Hidden	GRU acc	GRU params	GRU s/ep	LSTM acc	LSTM params	LSTM s/ep
16	0.8646	1,398	0.194	0.8515	1,830	0.202
32	0.8884	4,326	0.193	0.8711	5,702	0.202
64	0.8896	14,790	0.194	0.8973	19,590	0.199
128	0.8961	54,150	0.241	0.9083	71,942	0.369

The gated models beat the plain RNN by about 5 points. GRU is the cheapest per epoch at h=64 and has the lowest variance (std 0.0046). GRU accuracy saturates early (0.8884 at h=32 vs 0.8961 at h=128, 12.5x the parameters), while the LSTM keeps improving and reaches the best accuracy, 0.9083 at h=128, at 71,942 parameters and 0.369 s/epoch (1.8x the h=64 time). LSTM std is 0.009-0.025, so several differences are within seed noise.

(4) Wearable recommendation. GRU with hidden size 32: 0.8884 test accuracy with only 4,326 parameters (3.4x fewer than GRU-64 and 4.5x fewer than LSTM-64) at 0.193 s/epoch, within 0.0012 of GRU-64, and the GRU is the only model that solved the long-range adding task at all lengths (T up to 400), which matters for slowly varying activity context. If memory allows, GRU-64 (0.8896, 14,790 params) is the safe choice; LSTM-128 is not recommended because its +0.01 accuracy costs 71,942 parameters and about twice the time.

Problem 2 - summary of results

Task A (HAR Transformer, 3 seeds): no positional encoding 0.9313 +- 0.0089 acc / 0.9313 F1 (407,046 params); sinusoidal 0.9104 +- 0.0091 / 0.9106 (407,046); learned 0.9092 +- 0.0094 / 0.9090 (439,814). Time ~0.47-0.49 s/epoch for all. Positional information did not help here: the patch embedding of 8 raw samples already carries local order and the CLS classifier relies on the activity's overall signal statistics; the 0.02 gap is about two standard deviations, so it is a modest effect. Note the Transformer (0.93) is above the best Problem 1 model (0.8973 LSTM-64).

Task B (Tiny Shakespeare causal LM, seed 11): 1,115,394 characters, vocab 65, uniform guess 4.174 nats. Final validation loss 1.6753 nats = 2.4170 bits/char after 12x250 steps. Samples: temperature 0.2 is repetitive ('the shall son ... of the stand of the consul'), 0.8 is the most Shakespeare-like with invented words, 1.5 is noisy with broken words.

Experiment	Val loss (nats)	Bits/char
1 head (d_head 128)	1.9703	2.8425
4 heads (d_head 32)	1.9859	2.8651
8 heads (d_head 16)	1.9952	2.8785

Heads (4 epochs, equal d=128): fewer heads were slightly better in this short budget (1.9703 vs 1.9952); the differences are small (0.025 nats) and from one seed, so no strong conclusion. Attention heatmaps (last layer) show head 1 with a strong near-diagonal, local pattern and head 2 attending to a few earlier positions such as the newline/colon, i.e. heads specialise.

Causal mask removal: without the mask training loss collapses to 0.0773 and validation to 0.0246 nats (0.035 bpc) because position t reads the target at t+1; evaluated with the causal mask the same model gets 7.0032 nats (10.10 bpc), worse than the uniform guess 4.174, and its generation is degenerate ('ssssss sesese'). The masked model reaches 1.9859 and generates plausible text.

Context	Step time (s)	Peak memory (MB)	Time ratio vs 64	n^2 ratio
64	0.0104	86.23	1.00	1
128	0.0102	133.85	0.98	4
256	0.0188	267.46	1.81	16
512	0.0532	712.46	5.11	64

Scaling: attention scores grow as n^2 (262,144 to 16,777,216 per layer) and peak memory rises from 86 to 712 MB (8.3x for 8x the context, plus fixed model memory). Measured time has log-log slope 0.79 - below 2 because at small contexts the GPU is under-utilised (fixed launch overhead dominates); the cost becomes clearly super-linear at 256-512 (5.11x time for 8x context). The n^2 term dominates at longer contexts.